5.3.2

AI25BTECH11008 - Chiruvella Harshith Sharan

Question:

The pair of linear equations

$$\frac{3x}{2} + \frac{5y}{3} = 7 \quad \text{and} \quad 9x + 10y = 14$$

is

(a) consistent
(b) inconsistent
(c) consistent with one solution
(d) consistent with many solutions

Solution:

Step 1: Clear fractions and write augmented matrix

Multiply the first equation by 6 to clear fractions, obtaining $9x + 10y = 42$. We now have

$$9x + 10y = 42 \tag{1}$$
$$9x + 10y = 14 \tag{2}$$

The corresponding augmented matrix is

$$\left(\begin{array}{cc|c} 9 & 10 & 42 \\ 9 & 10 & 14 \end{array}\right). \tag{3}$$

Step 2: Row operation to check consistency

Perform the row operation $R_2 \leftarrow R_2 - R_1$ to eliminate the coefficients:

$$\left(\begin{array}{cc|c} 9 & 10 & 42 \\ 0 & 0 & -28 \end{array}\right). \tag{4}$$

The second row corresponds to $0x + 0y = -28$, which is impossible. Hence the system has no solution.

Final Answer:

$$\boxed{\text{The system is inconsistent.}}$$

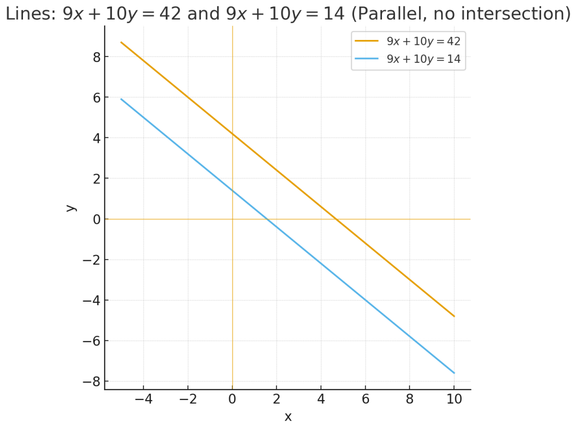

Fig. 1: Graph of the lines $9x + 10y = 42$ and $9x + 10y = 14$.